SIMATS ENGINEERING

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Course Code: **CSA0911** **Slot: A**

Course Name: **MACHINE LEARNING**

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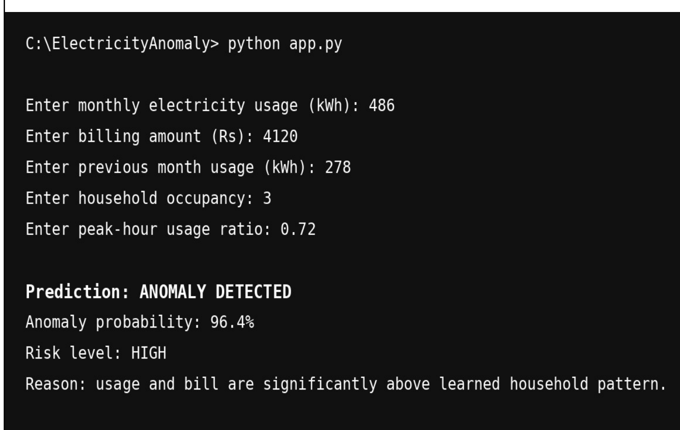
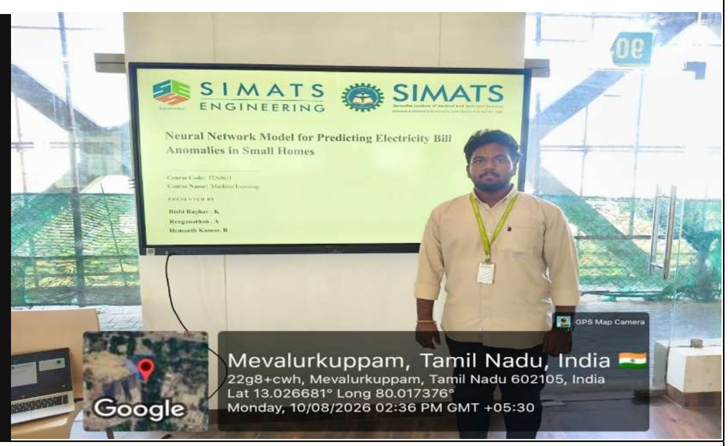
Project Title: NEURAL NETWORK MODEL FOR PREDICTING ELECTRICITY BILL ANOMALIES IN SMALL HOMES

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)





Project Description

The Neural Network Model for Predicting Electricity Bill Anomalies in Small Homes is a machine learning-based system designed to identify unusual or unexpected electricity bills by analyzing historical household electricity consumption and billing patterns. The system uses a Neural Network to learn normal electricity usage behavior and detect potential anomalies such as unexpectedly high or low bills.

The proposed system collects parameters such as monthly electricity consumption, previous bill amounts, number of occupants, appliance usage, billing period, seasonal variations, and historical consumption patterns. These factors are processed and provided to the neural network, which learns relationships between household characteristics and electricity consumption.

When new electricity consumption or billing data is entered, the trained model predicts the expected bill and compares it with the actual bill. If the difference exceeds a predefined threshold, the system identifies the bill as a potential anomaly. This can help users detect abnormal consumption, incorrect meter readings, billing errors, unusual appliance usage, or sudden changes in household electricity demand.

Student Signature **Guide Signature**